Task 1: Problem Definition and Dataset Selection

Problem Statement:

Credit card fraud to this day presents a persistent and costly issue for financial systems around the world. As digital and card not present transitions (online shopping, phone orders, subscription payments) have become the norm, chances for fraudulent activity has grown alongside them, and traditional rule-based detection systems (criteria list, blacklists, human review) struggle to keep up with both the ever-growing number of transactions and the constant adaptation of hackers. This project addresses a binary classification problem: given a set of transaction-level features, predict whether an individual credit card transaction is fraudulent or legitimate. The goal is to demonstrate how a data-driven, machine learning-based approach can support automated, scalable fraud screening.

Motivation and Background:

Global financial systems process millions of transactions on the daily, making manual review of every transaction time consuming and prone to human error. Machine learning offers a way to automatically flag suspicious transactions in sub real time by learning patterns from data rather than relying on static rules. However, fraud detection poses a difficult modelling challenge: fraudulent transactions are extremely rare compared to legitimate ones, meaning the dataset is highly imbalanced, and the cost of different types of errors is uneven. A missed fraudulent transaction (false negative) results in financial lost, whilst an incorrectly flagged legitimate transaction (false positive) inconveniences a customer and can harm the customer-company trust. This makes the problem a strong contender for exploring not just model accuracy, but metrics that keep up with real-world costs, such as precision, recall, the trade off, and the constant attempt to keep up with an ever-changing environment.

Stakeholders:

- **Cardholders** – Directly affected by monetary loss and the inconvenience of false declines on legitimate purchases
- **Banks** – Hold financial liability for confirmed fraudulent transactions and face compliance obligations as well as risk reputational damage from improper fraud handling
- **Merchants** – Soak up costs through chargebacks and loss of goods or services already sourced
- **Payment networks** – Rely heavily on system-wide trust in the network to maintain usage

Practical Relevance:

Effective fraud detection systems reduce financial losses in payments whilst maintaining customer trust and convenience. The connection between catching fraud and avoiding false alarms is integral to designing a proper system. A model that's tuned solely for accuracy can perform misleadingly well by assuming legitimate for almost every transaction, given how rare fraud is in the data. This'll grab more attention to evaluation strategy in the later stages of the project and shows an actual industry relevant trade off that data driven fraud systems must navigate in practice.

Task 2: Data Acquisition, Inspection, and Documentation

Yaacoub: The dataset was loaded in Python using Pandas, containing of 284,807 records across 31 columns. All columns are numeric: 30 are float64 (Time, V1-V28, Amount), and the target Class is int64, using binary values of 0 (legitimate) and 1 (fraudulent). Time represents elapsed seconds since the first transaction (0 to 172,792, roughly 48 hours), not proper timestamps. V1-V28 are anonymised, PCA-transformed features. Their original variables were not disclosed for security, so no domain level interpretation is possible, and their near-zero means reflect PCA's zero centring property. Amount ranges from $0 to $25,691.16, with a mean of $88.35 but a standard deviation of $250.12, indicating a right-skewed distribution. Inspection found no missing values across any columns. 1,081 duplicate rows were found. Unique value counts supported this: Time showed 124,592 unique values, V1-V28 showed an identical 275,663, and Amount showed 32,767. The identical count across all 28 PCA components suggested duplicates spanned rows rather than individual features. Class distribution revealed severe imbalance: 284,315 legitimate transactions (99.83%) versus 492 fraudulent (0.17%), a factor central to later decisions. No negative amounts were found. Two records have Time = 0, consistent with being the dataset's first observations rather than invalid data. No inconsistent entries or formatting issues, given the dataset is numeric with no text or date fields.

Key Assumptions: Time is treated as elapsed seconds only, not time of day, unless done later. V1-V28 are treated as numerical predictors with no real world meaning. The 1,081 duplicates are assumed redundant rather than repeat transactions, given no ID field exists to confirm otherwise. This will be touched on in Task 3. Records with Time = 0 are treated as valid.

Matthew:

Task 3: Data Cleansing and Transformation

Yaacoub: Of the checks from Task 2, missing values, invalid values, inconsistent categories, and formatting issues needed no action, since none were found. The 1,081 duplicate rows were identified earlier and addressed and by examining their class distribution, it showed 32 fraud cases and 1,790 legitimate cases among them. Since no transaction ID exists to confirm them, duplicates were removed to avoid data leakage between training and test data, reducing the dataset from 284,807 to 283,726 rows. Outliers in Amount were checked using the IQR method, which flagged 31,685 transactions (11.17%) as outliers. Given the high amount, this was taken as a sign the method doesn't work on Amount's right-skewed distribution, rather than evidence of invalid data. Checking the class distribution within these outliers showed 87 as fraudulent (0.275%) versus the datasets fraud rate of 0.17%, meaning fraud is slightly overrepresented among high value transactions. Given this, outliers were retained rather than removed. Instead, a log transformation (log1p) was used on Amount to address the right-skew, reducing skewness from 16.979 to 0.161 without removing data. Time and the log-transformed Amount were also standardised using StandardScaler, bringing both onto a comparable scale to the standardised PCA components (V1-V28). These cleaning and transformation steps were made into a single reusable function, clean_data(), which was confirmed to reproduce the manually cleaned dataset when applied to an untouched raw copy of the data. The final cleaned dataset has 283,726 rows and 31 columns.

Matthew:

Task 4: Exploratory Data Analysis and Visualisation

Yaacoub:

Class Distribution: The bar chart of Class shows an imbalanced target variable: legitimate transactions (Class = 0) outnumber fraudulent ones (Class = 1), consistent with the 99.83% vs 0.17% split shown in Task 2. On a linear scale, the fraud bar is barely visible; only when plotted on a log scale does it become clear. This confirms that modelling approach used later must take the imbalance into account, since a model predicting legitimate for every transaction would still get 99.8% accuracy whilst being practically useless.

Feature Distribution: The Time_scaled histogram isn't flat or uniform. It shows peaks and dips across the range, forming a wave-like pattern rather than an even spread. This shows the 48-hour span of the dataset, where transaction volume rises and falls with time of day, giving us the peak and dip shape. The Amount_scaled histogram is concentrated toward the lower range (approx. -2 to 1), then falls off past 1. The expected shape for transaction amounts even after log-transformation and scaling, with purchases being small to moderate and less as the amount increases.

Transaction Amount by Class: The boxplot of Amount_scaled by Class shows a difference in distribution shape between the two classes. Legitimate transactions have a median close to 0 with a tighter IQR (approx. -0.76 to 0.73), but a long tail of high value outliers reaching up to 4.22. Fraudulent transactions show a lower median (approx. -0.47) but a wider IQR (approx. -1.48 to 0.92) and a lower maximum (2.72), with no extreme outliers past the whiskers. Fraud does not follow the same "mostly small, sometimes very large" pattern in legitimate spending.

Correlation Analysis: A correlation heatmap confirmed V1-V28 are mostly unrelated with one another, consistent with their PCA origin. Against Class, the strongest negative correlations were V17 (-0.313), V14 (-0.293), and V10 (-0.207), while the strongest positive correlation was V11 (0.149), followed by V4 (0.129) and V2 (0.085). Amount_scaled (-0.008) and Time_scaled (-0.012) both showed weak correlation with Class, suggesting they are weak linear predictors of fraud on their own, though the boxplot showed fraud still differs in distribution shape in a way linear correlation doesn't show.

Missing Value Check: A visual missing value heatmap of the final dataset confirmed no missing values across any feature, consistent with Task 2 findings of zero nulls across 31 columns.

Key Data Insights: This analysis showed findings that shape how the dataset should be modelled going forward. Severe class imbalance is a huge factor, with only 473 of 283,726 transactions (0.167%) fraudulent, needing evaluation metrics past accuracy and imbalance handling techniques. Time_scaled's peak and dip pattern reflects daily transaction cycle and might have predictive value, despite showing negligible correlation with fraud. Amount_scaled stays right-skewed even after transformation, consistent with spending behaviour, and fraud and legitimate transactions differ in amount distribution shape rather than just central tendency. A subset of PCA-derived features (V17, V14, V12, V10, V11, V4) carry most of the linear signal for fraud detection, suggesting feature selection could be worthwhile at the modelling stage. Data quality stays strong throughout, with Task 3 cleaning decisions reflected here. Overall, the dataset is suited to fraud classification, though the class imbalance and low interpretability of PCA-derived features are challenges to address in later stages.

Independent-Samples T-Test: Transaction Amount by Class:

Null hypothesis: there is no difference in mean transaction amount between legitimate and fraudulent transactions.

Alternative hypothesis: there is a difference. Significance level: a = 0.05. Welch’s t-test comparing Amount_scaled between legitimate (n = 283,253, mean = 0.0003) and fraudulent (n = 473, mean = -0.1908) transactions returned t = 3.11, p = 0.00197. Since p is below 0.05, the null hypothesis is rejected. There is significant evidence the mean transaction amount differs by class. However, given the sample size, this significance does not mean a large difference, a point also shown in the weak correlation (-0.008) found between Amount_scaled and Class.

Chi-Square Test of Independence: Transaction Amount Category vs Class:

Since the dataset has no categorical variables, Amount_scaled was binned into four quartile categories (Low, Medium, High, Very High).

Null hypothesis: fraud rate is independent of amount category

Alternative hypothesis: fraud rate depends on amount category. Significance level: a = 0.05. The contingency table showed 213 and 168 fraud cases in Low and Very High bins, versus only 47 and 45 in Medium and High bins, despite almost identical total transactions per bin (70,000-71,000). This gave us x^2=185.42, p < 0.0001, df = 3, rejecting the null hypothesis. Fraud rate is associated with amount category, concentrated at both extremes and not the middle range. This indicates association and not causation, and the bin boundaries were a modelling choice rather than grouping, so different binning could show different category-level patterns.

Matthew: